

SMOL: a database of spectra and matrices of all small graphs

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Abstract

Cospectral graphs, non-isomorphic graphs that share the spectrum of an associated matrix, are central to spectral graph theory: they delimit exactly what a given matrix can and cannot “hear” about a graph's structure. Yet researchers studying cospectrality routinely re-enumerate small graphs and re-derive their spectra, often with floating-point eigensolvers whose rounding error can silently split or merge cospectral classes. We present **SMOL** (Spectra and Matrices Of Little graphs), a database of all 12,293,434 simple undirected graphs on up to 10 vertices (connected and disconnected), together with the spectra and cospectrality classifications of **16 graph matrices**, including standard operators (adjacency, the three Laplacians) and several less-tabulated ones (non-backtracking, distance-based, eccentricity, Yoon, and non- k -cycling matrices). Cospectrality is computed **exactly** for fourteen of the sixteen matrices, from integer or rational characteristic polynomials rather than floating-point eigenvalues, eliminating a class of precision errors that we show causes prior float-based enumerations to mis-count cospectral families (only the two largest operators, the non- k -cycling matrices, remain numerical). SMOL is distributed as a relational database with a documented schema and a public query API, providing a reusable “House of Graphs for spectra” that lets others look up, filter, and download cospectral families instead of recomputing them.

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1 Background & Summary

The spectrum of a matrix associated with a graph encodes structural information, and the study of *which* structure each matrix encodes is the subject of spectral graph theory [2, 3]. A sharp way to probe a matrix's discriminating power is to ask when two non-isomorphic graphs are **cospectral** for it. The fraction of graphs that are determined by their spectrum, and the structure of the cospectral families that are not, has been tabulated for the adjacency and Laplacian matrices at small orders [4, 5], and the House of Graphs project [8] provides a general-purpose, searchable catalogue of graphs and invariants. No comparable, openly queryable resource exists that (i) covers the full census of small graphs, (ii) spans a broad family of matrices beyond adjacency and Laplacian, and (iii) guarantees that its cospectrality classification is exact rather than floating-point approximate.

SMOL fills this gap. It enumerates every simple undirected graph on $n = 1, \dots, 10$ vertices (Table 1) using the canonical generator **geng** from the **nauty** package [1], computes 16 matrix spectra per graph, and groups graphs into cospectral families per matrix. The 16 matrices were chosen to span (a) the classical operators whose cospectrality is well studied, giving a validation baseline, and (b) operators that are of active interest but for which no small-graph cospectrality census has been published, where SMOL contributes new tabulations.

The central methodological choice is **exactness**. Floating-point eigenvalue hashing, the obvious way to test cospectrality at scale, fails for graphs whose spectra are numerically close but not equal, and for matrices whose eigenvalues are ill-conditioned. We instead key cospectrality on the exact characteristic polynomial for fourteen of the sixteen matrices, including the two non-backtracking matrices, whose complex spectra make floating-point classification least reliable. We show in Section 4 that this is not a theoretical nicety: at $n = 10$ the exact computation both recovers true cospectral families that an 8-decimal floating-point classification splits into singletons, and removes spurious families that rounding wrongly created, so a float-based census mis-counts cospectrality.

The complete dataset (12.3 million graphs, 16 matrices) is provided as a relational database with a documented schema (Section 3) and archived under a persistent identifier (Zenodo, DOI 10.5281/zenodo.20794132), with all code in a public repository. It is also served through a live query interface (currently at <https://smol-graphs-db.fly.dev>; the current address is linked from the repository).

2 Methods

2.1 Graph enumeration

For each order n from 1 to 10 we generate one representative of every isomorphism class of simple undirected graph using **geng** from **nauty**[1], including disconnected graphs. Graphs are stored in **graph6** format, the canonical ASCII encoding produced by **nauty**, which serves as the primary key. The census sizes are listed in Table 1 and match the known counts (OEIS A000088, [9]).

2.2 Matrices

SMOL computes spectra for the 16 matrix types in Table 2. For a graph G with adjacency matrix A and degree-diagonal D , we form:

- **Adjacency** A .
- **Kirchhoff (combinatorial) Laplacian** $L = D - A$.
- **Signless Laplacian** $Q = D + A$.
- **Normalized Laplacian** $I - D^{-1/2}AD^{-1/2}$.
- **Non-backtracking (Hashimoto) matrix** B on directed edges, and the **non-backtracking Laplacian** $I - D^{-1}B$ (complex spectra) [7].
- **Distance matrix** and its **Laplacian** ($\text{Tr} - \text{Dist}$), **signless Laplacian** ($\text{Tr} + \text{Dist}$), and **normalized** variant, where Tr is the diagonal of transmissions; defined for connected graphs only.
- **Eccentricity matrix** (connected graphs only) [12].
- **Yoon m -Laplacians** for $m = 2, 3$ [13].
- **Non- k -cycling** matrices for $k = 3, 4$ (complex spectra) [14].
- **k -blocking family**, a composite, eigenvalue-free *signature* over the k -blocking operators M_k for $k = 2, \dots, \Delta$.

Full definitions of the less-standard matrices are given in the cited sources and in the project glossary.

2.3 Spectral hashing and exact cospectrality

Two graphs are cospectral for a matrix if and only if they share its multiset of eigenvalues, equivalently if and only if they share its characteristic polynomial. SMOL assigns each (graph, matrix) pair a 16-character **spectral hash** and groups graphs with equal hashes into cospectral families. The hash is computed in one of two ways.

1. **Exact (13 matrices: adjacency, Kirchhoff, signless, normalized Laplacian, distance, distance Laplacian, distance signless, normalized distance, eccentricity, Yoon-2, Yoon-3, non-backtracking, and non-backtracking Laplacian).** We compute the exact characteristic-polynomial coefficients and hash those integers. For integer matrices this is the fraction-free Bareiss determinant of $xI - M$; for the normalized matrices it is a monic generalized characteristic polynomial of the underlying pencil; for the Yoon matrices the rational matrix is scaled to clear a common denominator before the integer computation. The two non-backtracking matrices have complex spectra but are indexed by directed edges, so they are at most $2m \leq 90$ dimensional for $m \leq 45$, small enough for an exact characteristic polynomial (the Hashimoto matrix is integer; the non-backtracking Laplacian $I - D^{-1}B$ is rational, computed exactly and reduced to a canonical integer coefficient tuple). Because the coefficients are exact, equal hashes mean genuinely equal spectra: no false splits or merges from rounding.
2. **Floating-point (2 matrices: non-3-cycling, non-4-cycling).** These have complex spectra and matrices of up to several thousand dimensions (non-4-cycling reaches

3024×3024 at $n = 9$) for which an exact characteristic polynomial is computationally infeasible. Their spectra are computed numerically (eigenvalues rounded to 8 decimals) and hashed. The k -blocking family is a composite signature over several operators and is treated separately.

This split is recorded per matrix in the schema (Table 2) so that users can see exactly which classifications are exact and which are numerical.

2.4 Storage and access

Spectra need not be stored: every eigenvalue array can be recomputed on demand from `graph6`, so SMOL stores only the 16 spectral hashes per graph plus structural invariants, keeping the database compact. Cospectral families (families of size at least two) are pre-computed into a dedicated table keyed by (matrix, n , hash); family members are recovered by an indexed lookup on the per-matrix hash column. The full database is PostgreSQL; a SQLite export drives a public query API.

3 Data Records

The dataset is archived at Zenodo (<https://doi.org/10.5281/zenodo.20794132>) under a **CC0 1.0 public-domain dedication** and comprises the relational database (`smol.db` and SQL dumps), per- n flat files (CSV) of the `graphs` and `cospectral_families` tables for users who do not want a database engine, the schema, and a data dictionary documenting every table and column.

The `graphs` table (one row per graph; 12,293,434 rows) holds `graph6` (primary key), n , m (edge count), the 16 `<matrix>_spectral_hash` columns, and structural invariants (bipartite, planar, and regular flags; diameter, radius, girth; minimum and maximum degree; triangle count; clique and chromatic numbers; and network-science descriptors such as algebraic connectivity and clustering). A NULL hash means the matrix is undefined for that graph (for example, distance-based matrices on disconnected graphs, or a nilpotent non- k -cycling operator).

The `cospectral_families` table (one row per cospectral family of size at least two) is keyed by (matrix_type, n , spectral_hash) with a family_size count. Members are the graphs sharing that (n , hash). Per-matrix totals are in Table 2.

Figure 3.1 shows, for each of the 16 matrices, the smallest cospectral family among connected graphs; Figure 3.2 shows the same restricted to graphs of minimum degree at least two. The non-backtracking Laplacian is the only matrix with no such family up to $n = 10$, since every minimum-degree ≥ 2 graph in the census is determined by its non-backtracking Laplacian spectrum.

Table 1. Census sizes (number of simple undirected graphs).

n	graphs
1	1
2	2
3	4
4	11

n	graphs
5	34
6	156
7	1,044
8	12,346
9	274,668
10	12,005,168
total	12,293,434

Table 2. The 16 matrices, classification method, and cospectral-family totals over all n (families = number of families of size at least two; members = graphs in some family; largest = size of the biggest single family).

key	matrix	spectrum	classification	families	members	largest
adj	Adjacency	real	exact	1,188,183	2,613,489	21
kirchhoff	Kirchhoff Laplacian	real	exact	677,396	1,456,934	23
signless	Signless Laplacian	real	exact	316,061	656,933	9
lap	Normalized Laplacian	real	exact	13,791	29,264	43
nb	Non-backtracking	complex	exact	229,247	2,074,004	398
nbl	Non-backtracking Laplacian	complex	exact	26,833	62,690	106
dist	Distance	real	exact	661,241	1,415,724	21
distlap	Distance Laplacian	real	exact	383,794	809,094	16
distsign	Distance Signless Laplacian	real	exact	159,568	327,991	9
distnorm	Normalized Distance Laplacian	real	exact	3,781	7,566	3
ecc	Eccentricity	real	exact	1,582,305	6,060,545	8,300
kblock_family	k -blocking family	signature	exact	99,556	263,920	580
yoona2	Yoon 2-Laplacian	real	exact	86	172	2
yoona3	Yoon 3-Laplacian	real	exact	10	20	2
non3cyc	Non-3-cycling	complex	numerical	63,559	204,415	242
non4cyc	Non-4-cycling	complex	numerical	14,084	48,308	141

4 Technical Validation

Reproduction of known counts. The census sizes (Table 1) match the known number of simple graphs (OEIS A000088, [9]). SMOL's count of graphs with an adjacency cospectral mate matches OEIS A006608 (the number of graphs not determined by their adjacency spectrum, [10]) exactly across the full range, *including* $n = 10$ (Table 3). The combinatorial-Laplacian and signless-Laplacian counts likewise reproduce the published Haemers-Spence tabulations [5].

Table 3. Graphs with an adjacency cospectral mate: SMOL versus OEIS A006608.

n	5	6	7	8	9	10
SMOL (exact)	2	10	110	1,722	51,039	2,560,606
A006608	2	10	110	1,722	51,039	2,560,606

n	5	6	7	8	9	10
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The $n = 10$ entry is the sharpest test, and it doubles as a demonstration of the exactness payoff: an 8-decimal floating-point classification gives 2,560,604 at $n = 10$ (two short), whereas the exact characteristic-polynomial count is 2,560,606, matching A006608. Exactness is what makes SMOL agree with the authoritative value.

Exactness recovers cospectral families that floating point misses. We compared, for each of the 13 exactly-classified matrices, the exact characteristic-polynomial classification against an 8-decimal floating-point eigenvalue classification. For the eleven real-spectrum matrices the two agree exactly at $n \leq 9$; at $n = 10$ the exact computation recovers a small number of cospectral families that rounding had split into singletons (Table 4), each a pair of graphs with genuinely equal spectra wrongly separated.

The two non-backtracking matrices, whose complex spectra are exactly where floating-point eigensolvers are least reliable, show the largest corrections. At $n = 10$, exact classification of the non-backtracking matrix removes 417 spurious cospectral families that 8-decimal rounding had created (by splitting true families inconsistently) and recovers 2,363 additional members from families rounding had split into singletons; the non-backtracking Laplacian gains one recovered family. We verified recovered families by independently recomputing the exact characteristic polynomial of their members; the exact classification never groups graphs with different characteristic polynomials, so no spurious merges occur.

Table 4. Cospectral families among the real-spectrum matrices at $n = 10$ recovered by exact computation that an 8-decimal floating-point classification splits (each recovered family is a pair).

matrix	recovered families	graphs
adjacency	1	2
distance Laplacian	2	4
eccentricity	4	8
signless Laplacian	1	2

Remaining numerical matrices. Only the two non- k -cycling matrices are classified numerically; their matrices reach several thousand dimensions, beyond exact characteristic-polynomial computation at this scale.

For the non-backtracking Laplacian, and for graphs of minimum degree at least two, SMOL's cospectral-family counts agree with the earlier enumeration of [7].

5 Usage Notes

The database can be queried three ways: (1) directly in SQL against the provided schema; (2) via the public REST API of the live interface (currently at <https://smol-graphs-db.fly.dev>, linked from the repository; with endpoints for single-graph lookup, filtered search, cospectral-family listing, and on-demand spectra); (3) by loading the per- n flat files. Eigenvalue arrays are not stored but are recomputed on demand from `graph6` (a single function call), so any spectrum in the dataset is reproducible bit-for-bit from the public code.

Smallest cospectral family per matrix (connected graphs)

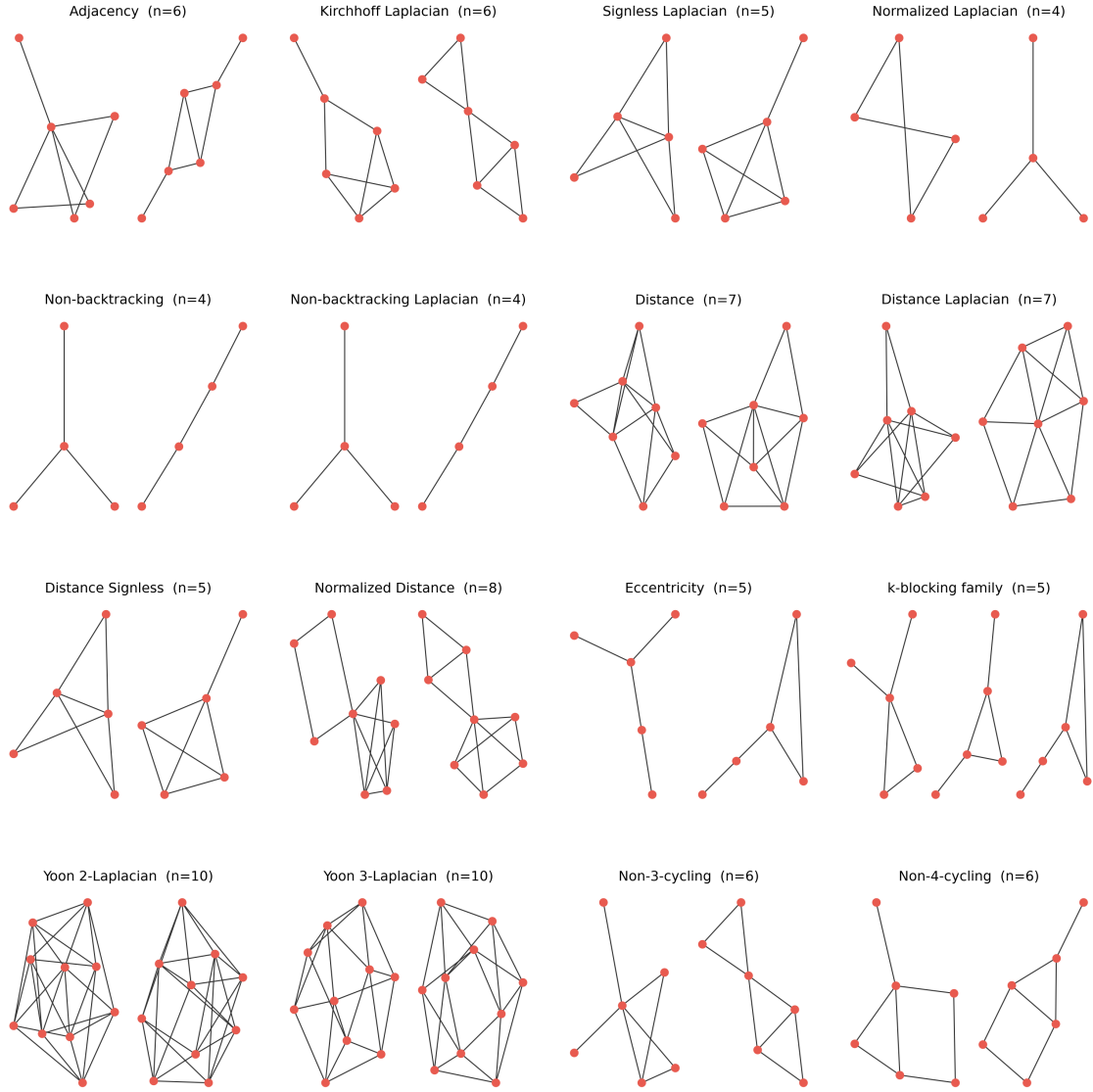


Figure 1. For each of the 16 matrices, the smallest cospectral family among connected graphs (the fewest-vertex family of non-isomorphic graphs sharing that matrix's spectrum); the number of vertices n is given per panel. The k -blocking family panel shows a triple; all others are pairs.

Smallest cospectral family per matrix (minimum degree ≥ 2)

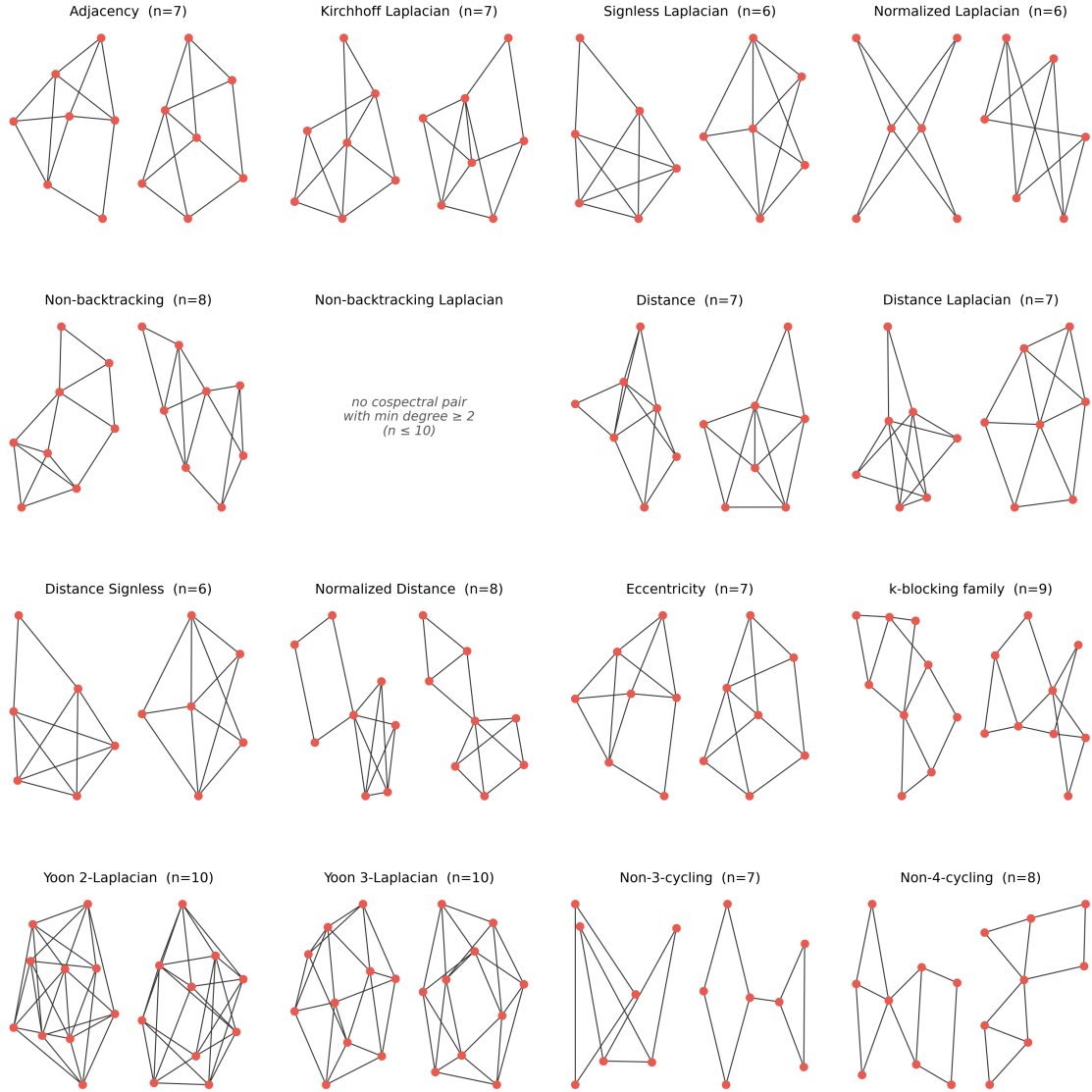


Figure 2. The smallest cospectral family per matrix among graphs of minimum degree at least two. The non-backtracking Laplacian has no such family up to $n = 10$, so every minimum-degree- ≥ 2 graph in the census is determined by its non-backtracking Laplacian spectrum.

Typical uses are looking up the cospectral mates of a given graph for any of the 16 matrices; enumerating all cospectral families of a chosen matrix and order to test a conjecture; or using the cospectral families as a benchmark of spectrally-indistinguishable graphs.

6 Code Availability

All code to regenerate the database (enumeration, matrix construction, exact characteristic-polynomial hashing, and the API) is openly available at <https://github.com/leotrs/smol> and archived with the dataset at Zenodo (<https://doi.org/10.5281/zenodo.20794132>) under the MIT license; the dataset itself is released under CC0 1.0. The generation pipeline is deterministic given `nauty`'s `geng`, so the entire dataset is reproducible from source.

7 Author Contributions

L.T. conceived the database, designed and implemented the generation pipeline and the exact characteristic-polynomial cospectrality hashing, performed all computations and validation, built the query interface, and wrote the manuscript.

8 Competing Interests

The author declares no competing interests.

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